

Big Data Analytics
CISELEC 501A-CS41S1

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Lab Activity-01: Understanding Data Characteristics and Quality

OBJECTIVE: To provide a hands-on experience in differentiating between various types of data and understanding their characteristics, identifying and addressing common data quality problems, and exploring dataset structures using real-world datasets. This activity will help students understand the importance of data quality and proper data understanding for effective data analysis and reliable IT systems.

Task 1: Introduction to Data and Datasets

1. Answer briefly the following key concepts

1. What are the three pillars of Big Data, and briefly describe each?

- Three pillars of Big Data are the 3 methodologies. CRISP-DM, SEMMA and Big Data Life Cycle.
CRISP-DM - The general way of creating a robust analytics
SEMMA - Must always starts at a sampling
Big Data Life Cycle - Centralizes at how it analyse other companies

2. Define what a Dataset is. What is an Attribute (also known as variable, field, characteristic, dimension, or feature), and how does it relate to an Object (also known as record, tuple, point, case, sample, entity, or instance)?

- A **dataset** is composed of datas and information in regards to a topic or a subject of a certain field. Attributes are those columns or identifiers for that value. It relates to an Object as per programmer's perspective is you can translate/transform that attribute into something that a machine can read.

3. List and describe the four main types of Attributes (Nominal, Ordinal, Interval, Ratio) with examples for each.

- **Nominal** - A representation of a value that is qualitative wherein it can be decided without order or ranking. (e.g.. {yes, no}, {red, blue, green})
- **Ordinal** - A representation of a value that is qualitative wherein it can be decided but with order or ranking (e.g.. {agree, neutral, disagree}, {high, mid, low})
- **Interval** - A representation of a value that is quantitative and has range but no true zero point.
(e.g.. {9:30am, 2am}time, {29degrees, 34degrees}temperature)
- **Ratio** - A representation of value that is quantitative wherein value has range and true zero point.
(e.g. {0cm, 24cm}height, {0kg, 8kg}mass {-500php, 1000php}debt)

4. Explain the four properties/operations (Distinctness, Order, Meaningful Differences, Meaningful Ratios) that define these attribute types.
 - **Distinctness** - A set of values that are unique to one another. This set of values are meaningful to **nominal** values.
 - **Order** - A set of values that has a ranking whether one of the values in a set is greater or lesser than the other values. This set of values are meaningful to **ordinal** values.
 - **Meaningful Differences** - A set of values wherein the difference of a value in a set relates to the other values. This set of values are meaningful to **interval** values
 - **Meaningful Ratios** - A set of values wherein the value is a multiplier of the other values. This set of values are meaningful to **ratio** values.
5. Differentiate between Discrete and Continuous Attributes, providing an example for each.
 - **Discrete** - Values that are distinct and can be counted (e.g. number of customer)
 - **Continuous** - Values that are distinct and can be counted or even predicted but it needs to be measurable. (e.g person's temperature)
6. Identify at least five common data quality problems discussed in the lecture.

2. Take notes on the following data structures and characteristics

- Data Matrix: Can any of the provided datasets be represented as a data matrix? Explain why or why not (3pts).
- One of the datasets provided is the Countries.csv that can be represented as a data matrix.

country	population	EU	coastline
Germany	80.68	yes	yes
United Kingdom	65.11	yes	yes
France	64.67	yes	yes
Italy	59.8	yes	yes
Spain	46.06	yes	yes
Poland	38.59	yes	yes
Romania	19.37	yes	yes
Netherlands	16.98	yes	yes
Belgium	11.37	yes	yes
Greece	10.92	yes	yes
Czech Republic	10.55	yes	no
Portugal	10.3	yes	yes

rows from the country is unique to one another.
 rows from the eu acts as a flag.

- Attribute Identification: For each of the "Countries.csv", "Cities.csv", "Players.xlsx", and "Teams.xlsx" datasets, identify at least three different attributes and categorize them (Nominal, Ordinal, Interval, Ratio, Discrete, Continuous) based on their properties (3pts).

- Cities.csv

city	country	country population	latitude	longitude	position	temperature
Kiruna	Sweden	9.85	67.85	20.22	east	-2.2
Bodo	Norway	5.27	67.25	14.4	east	4.5
Oulu	Finland	5.52	65	25.47	east	1.45
Trondheim	Norway	5.27	63.42	10.42	east	4.53
Abisko	Sweden	9.85	63.35	18.83	east	0.2
Tampere	Finland	5.52	61.5	23.75	east	3.59

position - **Nominal**

temperature - **Interval**

latitude - **Ratio**

- Countries.csv

country	population	EU	coastline
Germany	80.68	yes	yes
United Kingdom	65.11	yes	yes
France	64.67	yes	yes
Italy	59.8	yes	yes
Spain	46.06	yes	yes

eu - **Nominal**

coastline - **Nominal**

population - **Ratio**

- Players.xlsx

surname	team	position	minutes	shots	passes	tackles	saves
Abdoun	Algeria	midfielder	16	0	6	0	0
Belhadj	Algeria	defender	270	1	146	8	0
Boudebouz	Algeria	midfielder	74	3	28	1	0
Bougherra	Algeria	defender	270	1	89	11	0
Chaouchi	Algeria	goalkeeper	90	0	17	0	2
Djebbour	Algeria	forward	123	3	19	1	0
Ghezal	Algeria	forward	40	3	8	0	0
Guedioura	Algeria	midfielder	38	0	18	1	0
Halliche	Algeria	defender	270	2	94	4	0
Kadir	Algeria	midfielder	262	0	104	3	0

minutes - **Ratio**

passes - **Ratio**

tackles - **Ratio**

- Teams.xlsx

team	ranking	games	wins	draws	losses	goalsFor	goalsAgainst	yellowCards	redCards
Algeria	30	3	0	1	2	0	2	4	2
Argentina	7	5	4	0	1	10	6	7	0
Australia	20	3	1	1	1	3	6	7	2
Brazil	1	5	3	1	1	9	4	7	2
Cameroon	19	3	0	0	3	2	5	5	0
Chile	18	4	2	0	2	3	5	13	1
Denmark	36	3	1	0	2	3	6	6	0
Ecuador	8	4	4	2	4	2	5	6	0

ranking - **Ordinal**

redCards - **Ratio**

yellowCards - **Ratio**

- Data Quality Issues: Examine the provided datasets for potential data quality problems such as missing values, noise, outliers, or duplicate data. Note down any observations (3pts).
- One of the potential data quality problems I see is the countries and cities csv. The population is on decimal value where a population means that the number of people residing. There are no half people.

3. Summarize your understanding

- What is the purpose of understanding different data types and dataset structures in computer science?
- What is the role of identifying data quality problems in achieving reliable data analysis?
- How do the properties of attribute values (distinctness, order, differences, ratios) influence the appropriate analytical operations you can perform on data?
- One thing that I understand fully is that there are attributes property you can assigned to an attribute on a dataset. You can actually use this for you to analyse and make your own computation or algorithm by simply knowing where do these attributes belong.

Task 2: Exploring Data Quality Principles and Practical Application

1. Answer the following questions

- Why is it important to classify attributes (e.g., Nominal, Ratio) before beginning any analysis?
 - It is highly important to classify these attributes to know what kind of values you are dealing with on an attribute. Without knowing this, you will probably be lost at identifying what the values are supposed to be.
- How does understanding and addressing data quality issues (e.g., duplicates, outliers) contribute to more reliable and trustworthy results in your analysis?
 - Understanding and addressing data quality issues ensures a great margin and reduces margin errors when you are calculating your datas. Without resolving these issues, you will probably encounter a huge amount of gaps with the margins and other sectors.
- How does systematically applying this workflow help in streamlining data processing and reducing potential errors when performing calculations or creating charts?
 - By following the workflow above, it can streamline the data and reduce potential errors by making a way for a machine to compute and interpret these values in which they can fully understand. Since a computer knows well how to compute if given the right data.
- How would this workflow ensure better user confidence in the insights derived from the data?

- By showing how well the output relates to the given values in a dataset. Just by knowing that it is close to what you are expecting makes a huge difference to a user's morale. Or by simply realizing how to tweak it more.
2. **Write a short reflection (around 150 words)** : summarizing your chosen conceptual workflow's advantages in solving data-related challenges and producing reliable insights from datasets (10pts)
- The conceptual workflow advantage I chose is using a CRISP-DM methodology. Using this is enough for me to get started as someone with little experience. Following this workflow, can help me figure out more on how to analyze a dataset, preprocess and understand the dataset as a whole. Enabling me to produce a somewhat insightful dataset.

Case Study: Enhancing Data Reliability for Sports Analytics and Demographics (20pts)

A university research team is compiling a comprehensive database on global sports performance and demographic data to identify correlations between national characteristics and athletic success. Their current data sources ("Countries.csv", "Cities.csv", "Players.xlsx", and "Teams.xlsx") are not consistently categorized or validated, leading to potential misinterpretations in their research findings (e.g., miscounting players, incorrect averages for countries). By using a structured approach to understanding data characteristics and ensuring data quality, how could this research team improve the reliability of their findings and provide more accurate insights?

1. Key Considerations:

- **Data Integrity:** How do you maintain data accuracy and security during data integration and analysis, especially when combining attributes like country from different files?
 - One of the things we have to do is maintain data accuracy and securing by making sure that there is a key that connects between files or datasets. Making sure that the values are relatable. We can use the four properties of an attribute to identify this.
- **Scalability:** How can the approach to data understanding and quality outlined in this lab be applied to handle much larger datasets (e.g., data from all sports, all countries) that might overwhelm a spreadsheet (e.g., MS Excel)?
 - We can use a database for that. We should also consider different works such as Big Data Life Cycle. This is more appropriate methodologies to use specially for larger datasets. This is being used by large organizations such as Google and Amazon.

- **User Experience:** How will a properly structured and cleaned dataset (resulting from understanding data types and quality) impact the efficiency and confidence of the research team members (users) from different analytical departments?
 - By properly structuring and cleaning a dataset, it impacts the efficiency and confidence greatly by allowing research team members from different departments to look at the dataset in a different way. Approaching it in a way that can easily be understood by them. Enabling them to think of a new approach and how to approach it, leading to more efficient results.
- **Future Trends:** How will emerging technologies like cloud computing, AI, and machine learning impact the approach to data quality assessment and analysis for such large and varied datasets in the future?
 - Emerging technologies like cloud computing, AI, and machine learning highly impact the approach to data quality. It is because those technologies have millions OR even more than a billions of parameters that are fed to those models. Resulting for a machine to think differently. If this keeps improving where models no longer need to be fed of information, then large datasets are nothing to them. It'll be a swift action to take.

" I affirm that I have not given or received any unauthorized help in
this assignment, and that this work is my own "

Kenneth Arias

